



Department of Computer Science and Engineering (CSE)

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Course Code : EEE 2124

Course Name : Electronics Laboratory

Experiment No: 01

Experiment Name: Regenerating Diode I-V Graph

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Introduction:

In this experiment, we studied the I–V curve of a diode. The I–V curve is a graph that shows the relationship between voltage (V) and current (I) in a component. For a diode, this curve is not a straight line like a resistor — it has a unique shape.

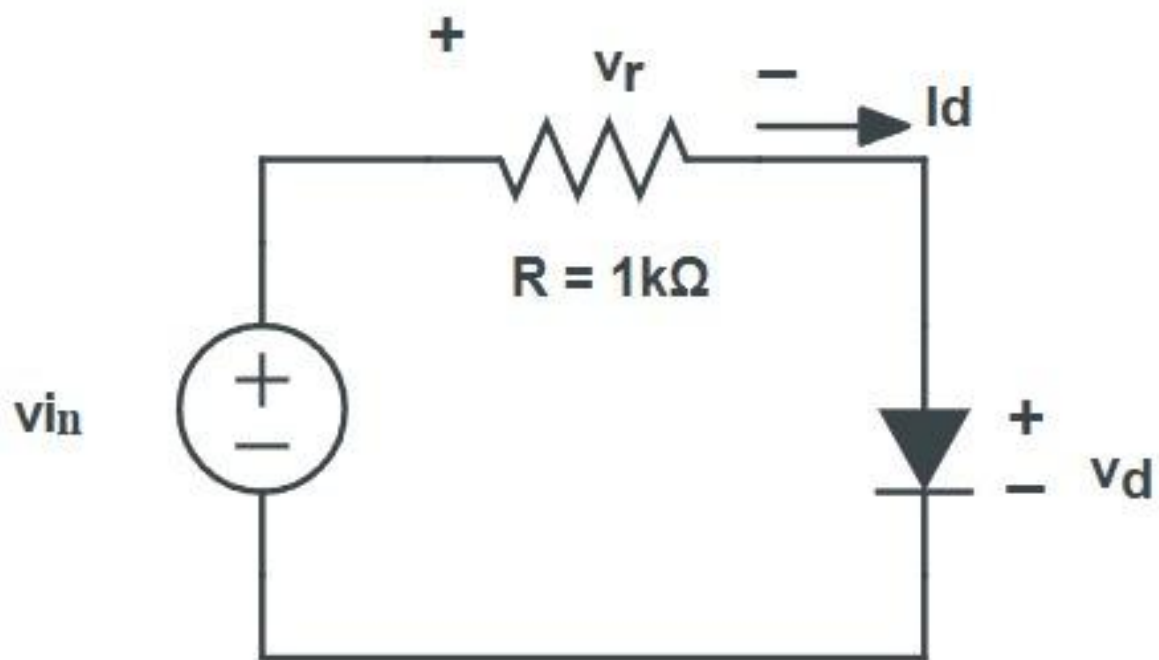
A diode allows current to pass only in one direction — this is called forward bias. It blocks current in the opposite direction (reverse bias). In forward bias, current only starts to flow properly after a certain voltage is reached. This is called the threshold voltage (around 0.7V for a silicon diode).

This experiment helps us see how the diode behaves when different voltages are applied, and how the current increases with it.

Equipments:

- DC Power Supply
- Multimeter
- Bread Board
- Wires
- Resister
- Capacitor
- Diode
- Oscilloscope
- Function Generator

Circuit Diagram:



Data Collection:

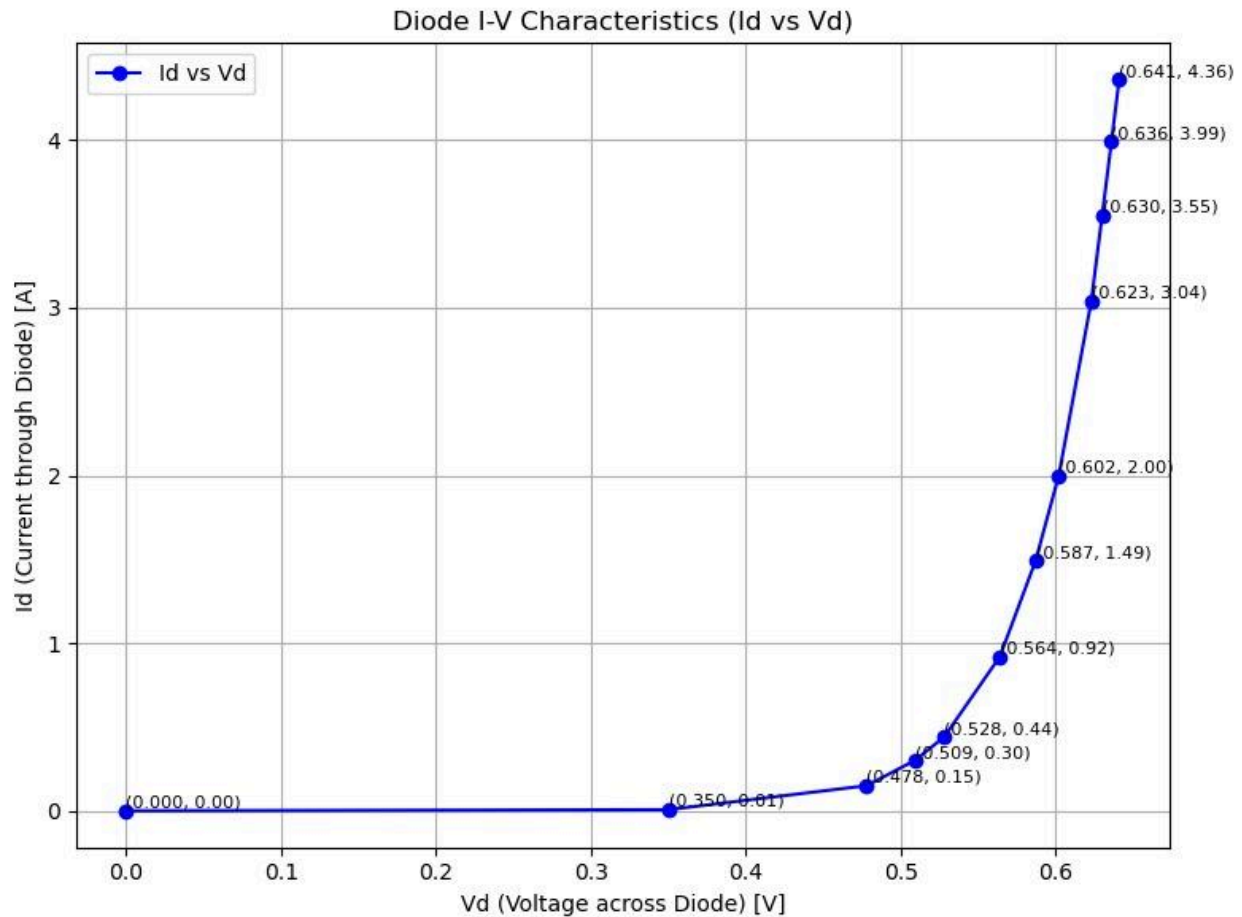
Measured Resistance, $R = 0.988 \text{ k}\Omega$

Formula Used:

$$I_d = \frac{V_r}{R} = \frac{V_r}{0.988}$$

Vin (Voltage Input)	Vr(Voltage across Resistor)	Vd (Voltage across Diode)	$I_d = \frac{V_r}{R}$
0	0	0	0
0.4	0.007	0.35	7.08×10^{-3}
0.6	0.148	0.478	0.15
0.8	0.288	0.509	0.3
1	0.438	0.528	0.44
1.5	0.910	0.564	0.92
2.1	1.47	0.587	1.49
2.6	1.973	0.602	2
3.6	3.003	0.623	3.04
4.1	3.505	0.630	3.55
4.6	3.941	0.636	3.99
5	4.31	0.641	4.36

Diode I-V Graph:



Discussion:

This experiment clearly shows how a diode behaves under forward bias. At first, even if we increase the voltage slowly, the current doesn't rise much. But once the voltage crosses about 0.7V, the current starts increasing rapidly. This is because silicon diodes need a minimum voltage to start conducting — that's their threshold.

Key takeaways:

- Diodes allow current to flow only after a certain voltage.
- The I–V graph helps us see how the diode turns "ON" gradually and then fully.
- This behavior is very different from a resistor, which has a straight-line (linear) graph.

This experiment is useful because it helps us:

- Understand the working principle of a diode
- Learn how to read and create I–V graphs
- Practice using multimeters and power supplies in real circuits

This type of graph is not just academic — it's used by engineers to test and verify components in real-world devices like chargers, radios, TVs, and more.